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NumLab: A WEB APPLICATION FOR NUMERICAL ANALYSIS

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MATHEMATICS**

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Declaration

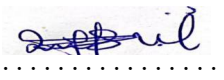
We hereby conscientiously declare that this dissertation is our own work under the able supervision of E Owusu-Ansah, towards the award of the Bachelor of Science(BSc.) in Mathematics and that, to the best of our knowledge, it contains no material previously published by another person nor material which had been accepted for the award of any other degree of the university, except where due acknowledgement had been made in the text.

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Dedication

We humbly dedicate this work to the Almighty God for His grace, mercy and favour throughout our four-year academic endurance. We are not self made, our reaching this point in our academic life has been by the collective effort of some family and friends and so we would like to dedicate this work to every single individual who has contributed invaluablely in one way or the other to our reaching this point in life.

Abstract

The aim of this project is to develop "NumLab," a web application designed to meet the needs of students worldwide by providing a centralized platform for learning and practicing numerical analysis. NumLab aims to change students' perceptions of numerical analysis as a difficult subject by offering a user-friendly interface that enhances their academic performance. Through intuitive interfaces and dependable backend systems, the application offers a reliable and convenient solution for mastering numerical analysis. This documentation outlines the objectives, design, development, and testing processes involved in creating NumLab.

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Introduction

1.1 Background of the Study

Our thesis focuses on developing a web application designed to facilitate the easy solving of numerical analysis problems. The name NumLab, which stands for "Numerical Laboratory," denotes a systematic approach to addressing mathematical issues and accomplishing analytical goals. The application NumLab is derived from "Numerical Laboratory," indicating its purpose as an extensive tool for performing and researching numerical analysis. NumLab is primarily designed for students and professionals, including engineers, scientific computing experts, data analysts, statisticians, economists, and those involved in financial modeling. Numerical analysis, a branch of mathematics, employs numerical methods to solve mathematical problems. These methods are essential for a wide range of applications across various fields. Despite its importance, numerical analysis is often perceived as challenging and complex, which can hinder the learning process. Our web application aims to address this issue by providing a user-friendly platform that simplifies numerical analysis, making it more accessible and manageable for users. Through NumLab, we strive to enhance the understanding and application of numerical analysis, ultimately improving academic and professional performance in related fields.

1.2 Problem Statement

Numerical analysis is a fundamental aspect of many academic disciplines, yet it is often perceived as challenging and inaccessible by students worldwide. This negative perception can hinder students' academic performance and their willingness to engage with the subject. The lack of a centralized, user-friendly platform for learning and practicing numerical analysis intensifies this issue, making it difficult for students to find reliable resources and support. There is a pressing need for an intuitive, comprehensive web application that can provide students with the resources and support they need to overcome these challenges and succeed in their studies.

1.3 Project Aim

The aim of this project is to develop "NumLab," a comprehensive web application designed to support students worldwide in mastering numerical analysis. By providing a centralized platform with user-friendly interfaces and dependable backend systems, NumLab seeks to transform students' perceptions of numerical analysis from a challenging subject to an accessible and manageable one. The application aims to enhance academic performance by offering intuitive tools, reliable resources, and guided practice, making the learning process more efficient and effective.

1.4 Objectives

1. **Develop a User-Friendly Interface:** Design an intuitive and accessible user interface that simplifies the process of learning and practicing numerical analysis.
2. **Provide Comprehensive Learning Resources:** Offer a wide range of learning materials, including tutorials, examples, and interactive exercises, to support students in mastering numerical analysis.
3. **Implement Reliable Backend Systems:** Ensure the backend systems are robust and dependable to provide seamless performance and reliability.
4. **Enhance Academic Performance:** Improve students' understanding and application of numerical analysis through guided practice and feedback.
5. **Expand Accessibility:** Make the platform available to students and professionals globally, regardless of their location.

1.5 Scope of the Project

The scope of this project includes the following:

1. **Target Audience:** The primary users of NumLab will be students and professionals in fields such as engineering, scientific computing, data analysis, statistics, economics, and financial modeling.
2. **Functionality:** The web application will include features such as interactive exercises, problem-solving tools, forum, formula comparison.
3. **Geographical Reach:** The application will be designed to be accessible to users worldwide, with considerations for multilingual support and different educational curricula.
4. **Usability Testing:** The project will involve rigorous testing to ensure the application is user-friendly, reliable, and effective in achieving its objectives.

1.6 Approach to Project

1. **Research and Planning:** Conduct thorough research on the needs of students and professionals in numerical analysis and plan the features and design of the web application accordingly.
2. **Design and Development:** Create wireframes and prototypes for the user interface, followed by the development of the frontend and backend systems using appropriate technologies.
3. **Content Creation:** Develop comprehensive learning materials, including tutorials, examples, and interactive exercises.
4. **Testing and Feedback:** Conduct usability testing with a diverse group of users to gather feedback and make necessary improvements.

5. **Deployment and Maintenance:** Deploy the application on a reliable hosting platform and ensure ongoing maintenance and updates to keep the application functional and up-to-date.

Literature Review

2.1 Background Review

Lecture notes, slides, and books have long been essential in helping students perform well. These systems have evolved from manual to advanced formats. Initially, students had to write notes and go through numerous pages in books before grasping a particular concept. Students often struggle when their notes are missing or when they mistakenly delete slides sent by their lecturers or tutors. Additionally, not having access to a specific book for a lecture can be a significant obstacle. Despite attempts to find a solution, existing systems are not perfect and have room for improvement.

2.2 Review of Existing Application

AtozMath

A wide variety of mathematical tools, resources, and educational materials can be found on the internet platform AtoZMath. Its goal is to help professionals, teachers, and students solve a variety of challenging mathematical problems and improve their comprehension of mathematical ideas.

Advantages

1. Comprehensive Resource:

- (a) Wide Range of tools: AtoZMath offers a wide range of mathematical tools, such as graphing calculators, problem solvers, and specialised calculators for various mathematical fields.
- (b) Educational Content: From basic to advanced, the platform provides interactive lessons, practice tasks, and tutorials to meet different learning needs.

2. Interface Friendly:

- (a) Intuitive Design: Designed with user accessibility and ease of use in mind, the platform works with users of all ages and technical skill levels.
- (b) Effective Navigation: Time is saved and the learning process is improved when users can quickly locate the tools and resources they require.

3. Accessibility:

- (a) Online Availability: AtoZMath is a handy tool for professionals, educators, and students on the go because it can be accessed from any location with an internet connection.
- (b) 24/7 Access: The portal is always open, offering users unceasing assistance.

4. Educational Support:

- (a) Assignment Assistance: AtoZMath provides explanations and solutions for textbook problems, assisting students with their assignments and strengthening their understanding of the subject matter.
- (b) Detailed Solutions: Rather than focusing only on the solution, step-by-step solutions assist users in understanding the process of problem-solving.

5. Enhanced Learning:

- (a) Visual Tools: Using graphing calculators and other visual aids, users can visualise difficult mathematical ideas to better grasp them.
- (b) Interactive Learning: Users are engaged and encouraged to learn effectively through interactive lessons and practice exercises.

Disadvantage

1. it relies on technology:

- (a) Internet Requirement: Since AtoZMath is an online programme, users with poor or no internet connectivity may find this to be a hindrance.
- (b) Device Compatibility: Depending on the device and browser being used, AtoZMath's accessibility and performance may change.

2. The standard of the content:

- (a) Accuracy and Verification: Although AtoZMath strives to offer accurate solutions, users should independently confirm the outcomes because mistakes or inaccuracies could happen.
- (b) Explanation Depth: Not all users will find all explanations to be extensive enough, especially those who require more thorough instruction or a deeper comprehension.

3. Learning Style Compatibility:

- (a) Passive Learning: If users only utilise the platform to get answers without really interacting with it, this could promote passive learning.

Wolfram/Alpha

Wolfram Research created the computational knowledge engine known as Wolfram—Alpha. It is frequently used to calculate answers to questions, carry out calculations, produce reports, and more.

Advantages

1. Comprehensive Knowledge Base: Wolfram—Alpha is an effective tool for discovering knowledge and solving problems in a variety of sectors since it has a large database covering a wide range of topics.
2. Accurate and Reliable: To produce results that are accurate and dependable, the engine makes use of validated data and complex algorithms.

3. Usability: Users may enter questions in normal language thanks to the user-friendly interface.
4. Educational Tool: It's a great resource for teachers and students alike because it provides step-by-step instructions and solutions.
5. Real-Time Computation: Wolfram—Alpha is helpful for urgent problem-solving needs because it can execute computations in real time.

Disadvantage

1. Complexity for Beginners: Wolfram—Alpha's scope and depth may be difficult for new users who may not know how to formulate questions or evaluate findings correctly.
2. Requires online Access: The majority of functions can't be used in offline circumstances because they depend on an online connection.
3. Limited Free Access: Although a lot of functions are free, a paid membership is frequently needed to access advanced features and comprehensive step-by-step answers.
4. Not a Replacement for Learning: Learning can be harmed by relying on Wolfram—Alpha for answers if the underlying concepts are not understood.
5. Limitations of Natural Language: While it accepts natural language input, complicated or poorly worded searches could not produce the intended outcomes.

Codesansar

A website called Codesansar offers resources to help with studying several programming languages and technologies, including source codes and tutorials. The website provides thorough explanations, sample programmes, and practice activities for a number of programming languages, including C, C++, Python, Java, and others. A helpful tool for learning and using numerical techniques is Codesansar. Numerical methods are widely utilised in many disciplines, including computer science, physics, engineering, and finance. They involve algorithms for solving mathematical problems numerically rather than analytically.

Advantages

1. No cost and availability:
Because Codesansar is open source, learning and using programming and numerical techniques can be done at a reasonable cost.
2. A Wide Range of Subjects:
The website offers a comprehensive learning resource, including a wide range of programming languages and numerical approaches.

3. Useful Illustrations:

The website provides a wide range of sample programmes and step-by-step tutorials that can help in understanding and efficient use of numerical approaches.

4. Engaging in Interactive Education:

Codesansar offers tasks, tests, and project ideas to help with active learning and concept reinforcement.

Disadvantages

1. Varying Content Quality:

The level and calibre of the tutorials and examples can differ, as is the case with many free resources. Certain explanations might not be very clear or in-depth.

2. Restricted Communication:

Online learning can be less participatory than traditional classroom or course settings, with fewer opportunities for in-person question-and-answer sessions.

3. Absence of Accreditation

One potential disadvantage for individuals seeking credentials is that learning from Codesansar does not result in any official certification or accreditation.

4. Self-control is Needed:

Some students may find it difficult to maintain the high levels of self-discipline and drive needed for self-paced study.

5. Restricted Advanced Subjects/Few Advanced Subjects:

Even though Codesansar covers a wide range of subjects, it might not go into very specific or complex numerical method areas that are the subject of more advanced textbooks or college courses.

Problem Identification

The reviewed systems for numerical analysis, including Codensar, Wolfram, and Atoz-math, have several flaws that need to be addressed, including:

1. Lack of comprehensive and user-friendly tutorials.
2. Limited accessibility across multiple platforms.
3. Absence of real-time collaboration features.
4. Basic or inefficient visualization tools for complex data.

Project Evaluation

In summary, existing numerical analysis tools face challenges such as limited platform accessibility, lack of real-time collaboration, and inadequate visualization tools. Addressing these limitations in NumLab can significantly improve the user experience and effectiveness of numerical analysis tools.

Methodology

Nowadays, one of the areas with greatest need for research is numerical analysis. Numerical techniques are used in order to easily acquire an almost exact answer for physical problems, as finding an exact solution is extremely complex and, again, for most physical problems, there is no exact solution at all.

NumLab is an application that helps users tackle numerical analysis problems.

With its easy-to-use interface, it provides access to a variety of numerical methods that fall into three main categories: numerical calculus, non-linear systems, and linear systems.

In order to facilitate learning and problem-solving, the program offers a wealth of resources and interactive features.

Landing Page

Users are presented with a landing page 3.1 when accessing NumLab, which provides a concise synopsis of the application's features and goal.

This tab is where you can access all of the features that are included in the app.

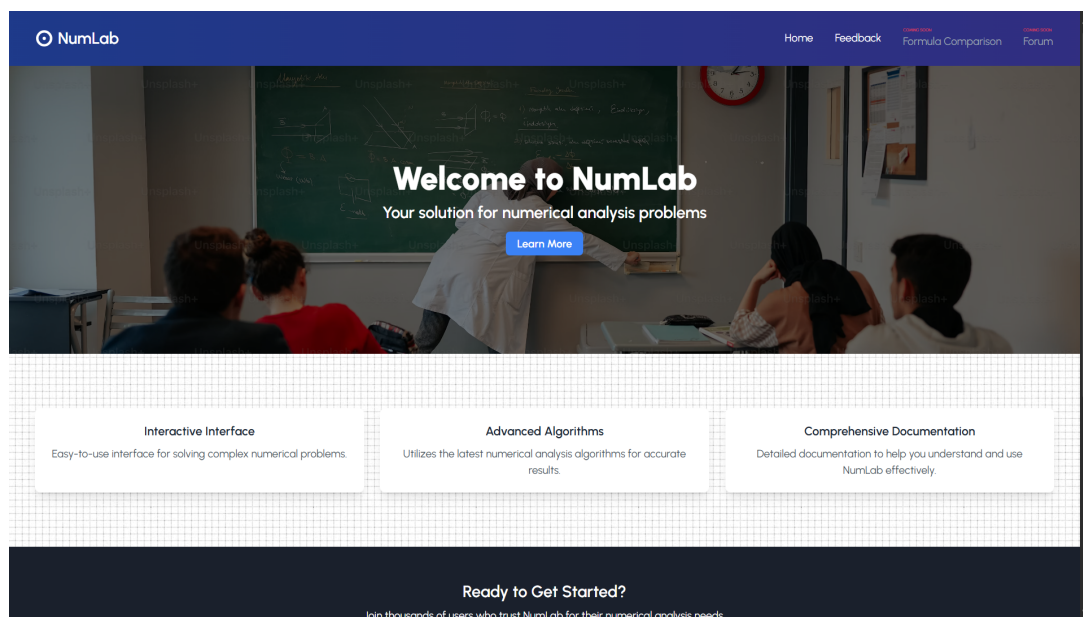


Figure 3.1: Landing Page

Navigation and Exploration

On the landing page, the **"Learn More"** button is easy to see. Clicking this button directs users to a page where they can review numerical analytic methods broken down into three categories:

1. **Non-Linear Systems:** This category comprises methods for resolving equations where the variables do not have a linear connection.

2. **Numerical Calculus:** Techniques for differentiating and integrating numerical data are covered in this section.
3. **Systems Linear:** This category includes methods for addressing problems involving systems of linear equations.

The figure 3.2 below shows the homepage of the various categories under numerical analysis;

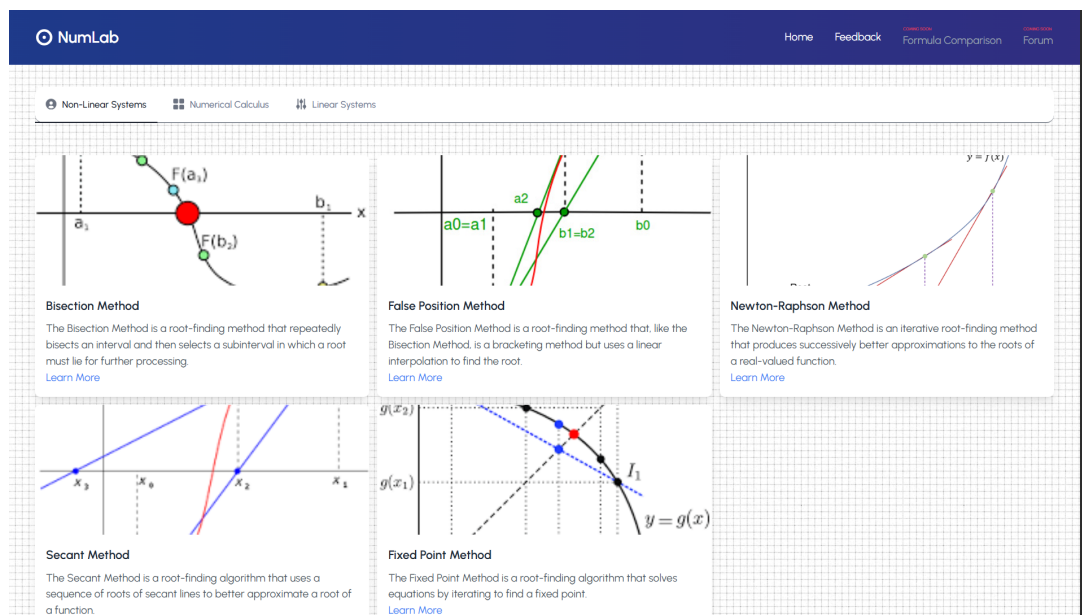


Figure 3.2: Home Page

Upon selecting a method from any of the categories, users are presented with the following resources:

1. **Background:** Provides an overview of the method, including its theoretical foundation and key concepts.

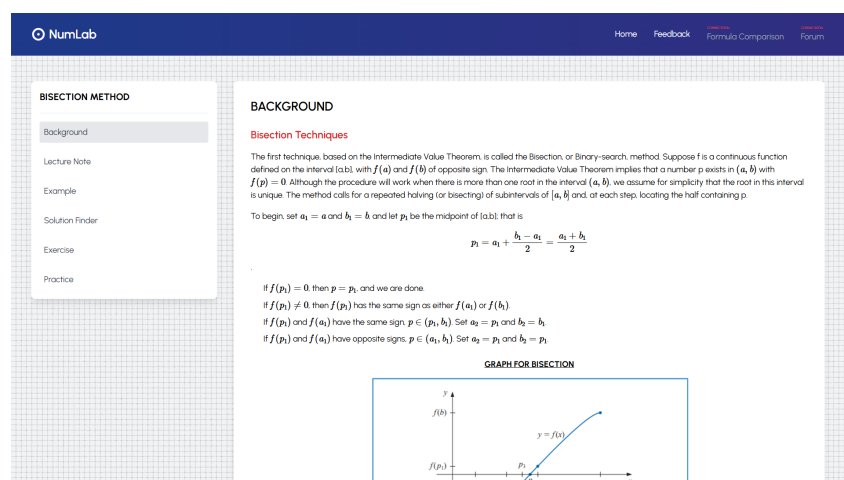


Figure 3.3: Background Overview

2. **Lecture Notes:** Offers detailed notes on the method, including derivations, applications, and theoretical explanations.

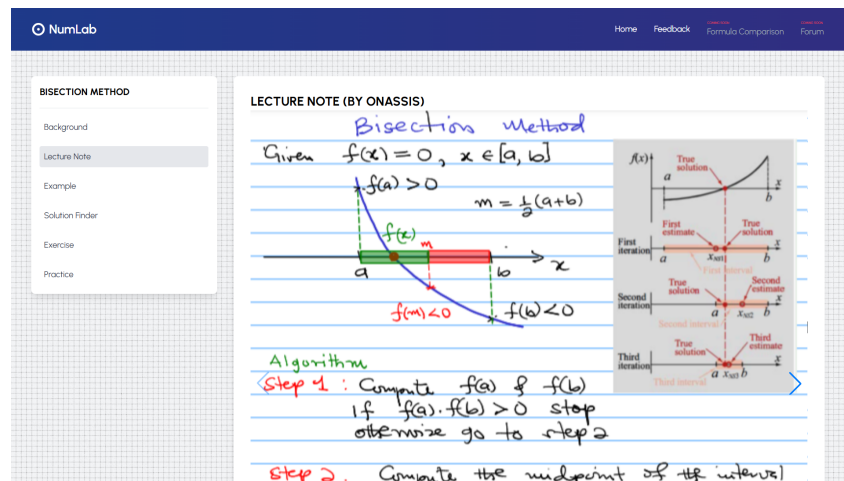


Figure 3.4: Lecture Notes

3. **Solution Finder:** This interactive tool allows users to input functions and other parameters to perform step-by-step calculations. After entering the required information, users click the “**Calculate**” button to view the detailed solution process.

Figure 3.5: Solution Finder Tool

4. **Example:** Presents a worked example demonstrating the application of the method to a specific problem.

BISECTION METHOD

Background
Lecture Note
Example
Solution Finder
Exercise
Practice

EXAMPLE

Use the bisection method to find the root of $f(x) = x^{10} - 1$, $x \in [0, 1.8]$

Solution
Let $f(x) = x^{10} - 1$
 $f(0) = 0^{10} - 1 = -1 < 0$ $f(1.8) > 0$
 $f(1.8) = 1.8^{10} - 1 = 12.7858 > 0$
Since $f(0) \cdot f(1.8) < 0$ it follows that $[0, 1.8] = [a, b]$ contains the root of $f(x)$.

1st Iteration
 $m = \frac{a+b}{2} = \frac{0+1.8}{2} = 0.65$
 $f(m) = f(0.65) = (0.65)^{10} - 1 = -0.98658 < 0$
 $f(a) < 0$ $f(m) < 0$ $f(b) > 0$
 $a = 0$ $m = 0.65$ $b = 1.8$
Since $f(m) \cdot f(b) < 0$, the new interval is $[a, b] = [m, b] = [0.65, 1.8]$

2nd Iteration
 $m = \frac{a+b}{2} = \frac{0.65+1.8}{2} = 0.975$

Figure 3.6: Worked Example

5. **Exercise:** Contains unsolved problems for users to practice and apply the method. These exercises are designed to reinforce understanding and proficiency.

BISECTION METHOD

Background
Lecture Note
Example
Solution Finder
Exercise
Practice

EXERCISE

- Use the Bisection method to find p_1 for $f(x) = \sqrt{x} - \cos x$ on $[0, 1]$.
- Let $f(x) = 3(x+1)(x-\frac{1}{2})(x-1)$. Use the Bisection method on the following intervals to find p_1 .
a. $[-2, 1.5]$ b. $[-1.25, 2.5]$
- Use the Bisection method to find solutions accurate to within 10^{-2} for $x^3 - 7x^2 + 14x - 6 = 0$ on each interval.
a. $[0, 1]$ b. $[1, 3.2]$ c. $[3.2, 4]$
- Use the Bisection method to find solutions accurate to within 10^{-2} for $x^4 - 2x^3 - 4x^2 + 4x + 4 = 0$ on each interval.
a. $[-2, -1]$ b. $[0, 2]$ c. $[2, 3]$ d. $[-1, 0]$
- Use the Bisection method to find solutions accurate to within 10^{-1} for the following problems.
a. $x - 2^{1/x} = 0$ for $0 \leq x \leq 1$
b. $x^2 - x^3 + 3x - 2 = 0$ for $0 \leq x \leq 1$
c. $2x \cos(2x) - (x+1)^2 = 0$ for $-3 \leq x \leq -2$ and $-1 \leq x \leq 0$
d. $x \cos x - 2x^2 + 3x - 1 = 0$ for $0.2 \leq x \leq 0.3$ and $1.2 \leq x \leq 1.3$
- Use the Bisection method to find solutions, accurate to within 10^{-3} for the following problems.
a. $3x - e^x = 0$ for $1 \leq x \leq 2$
b. $2x + 3 \cos x - e^x = 0$ for $0 \leq x \leq 1$
c. $x^3 - 4x + 4 - \ln x = 0$ for $1 \leq x \leq 2$ and $2 \leq x \leq 4$
d. $x + 1 - 2 \sin \pi x = 0$ for $0 \leq x \leq 0.5$ and $0.5 \leq x \leq 1$

NumLab Quick Links Non-Linear Systems Linear Systems Numerical Calculus
Email: support@numlab.com Non-Linear Systems Bisection Method Gaussian Elimination Method Finite Method

Figure 3.7: Exercise Problems

6. **Practice:** Includes a set of multiple-choice questions related to the method. Users are presented with five random questions at a time, with options to select and submit answers. Upon submission, users receive immediate feedback, including the correct answers.

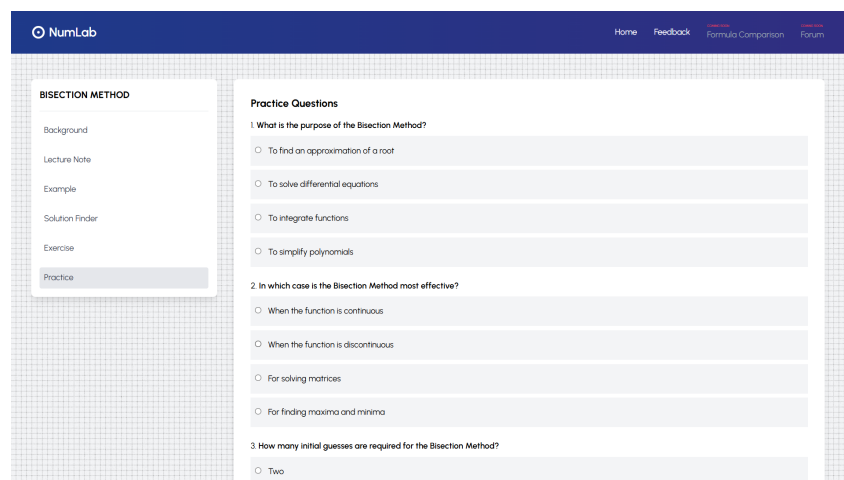


Figure 3.8: Practice Questions

3.1 Bisection Method

The Bisection method is a method used for finding roots of continuous functions. This process involves finding a root, or solution, of an equation of the form $f(x) = 0$, for a given function f . Suppose f is a continuous function defined on the interval $[a, b]$, with $f(a)$ and $f(b)$ of opposite sign. The Intermediate Value Theorem implies that a number p exists in (a, b) with $f(p) = 0$. Although the procedure will work when there is more than one root in the interval (a, b) , we assume for simplicity that the root in this interval is unique. Burden and Faires [2010] The method calls for a repeated halving (or bisecting) of subintervals of $[a, b]$ and, at each step, locating the half containing p . To begin, set $a_1 = a$ and $b_1 = b$, and let p_1 be the midpoint of $[a, b]$; that is,

$$p_1 = a_1 + \frac{b_1 - a_1}{2} = \frac{a_1 + b_1}{2}$$

1. If $f(p_1) = 0$, then $p = p_1$, and we are done.
2. If $f(p_1) \neq 0$, then $f(p_1)$ has the same sign as either $f(a_1)$ or $f(b_1)$.
3. If $f(p_1)$ and $f(a_1)$ have the same sign, $p \in (p_1, b_1)$. Set $a_2 = p_1$ and $b_2 = b_1$.
4. If $f(p_1)$ and $f(a_1)$ have opposite signs, $p \in (a_1, p_1)$. Set $a_2 = a_1$ and $b_2 = p_1$.

ALGORITHM FOR BISECTION METHOD (PYTHON)

```

from sympy import symbols, lambdify, parse_expr

def bisection_method(equation_str, a, b, tol=1e-6, max_iter=1000):
    iterations = 0
    data = []
    intervals = []

    x = symbols('x')
    equation = parse_expr(equation_str)
    equation = lambdify(x, equation)

    if equation(a) * equation(b) >= 0:
        raise ValueError("Bisection method fails.")

    while (b - a) / 2 > tol and iterations < max_iter:
        c = (a + b) / 2
        root_estimate = (a + b) / 2
        func_value = equation(root_estimate)
        data.append((iterations, root_estimate, func_value))

        # Save the current interval
        intervals.append((a, b))

        if func_value == 0:
            break
        if equation(c) * equation(a) < 0:
            b = root_estimate
        else:
            a = root_estimate
        iterations += 1

    return root_estimate, data, intervals

```

Convergence and Error of Bisection Method

Let $[a_0, b_0] = [a, b]$ be the initial interval, with $f(a)f(b) < 0$. Define the approximate root as $x_n = \frac{(b_{n-1} + a_{n+1})}{2}$. Then there exists a root $x^* \in [a, b]$ such that

$$|x_n - x^*| \leq \left(\frac{1}{2}\right)^n (b - a)$$

More over to achieve accuracy of $|x_n - x^*| \leq \epsilon$ it suffies to take

$$n \geq \frac{\log(b-a) - \log \epsilon}{\log 2}$$

Proof. It is obvious that

$$b_n - a_n = \frac{1}{2}(b_{n-1} - a_{n-1}),$$

which implies that

$$b_n - a_n = \left(\frac{1}{2}\right)^n (b_0 - a_0).$$

Therefore,

$$|x_n - x^*| \leq \frac{1}{2}(b_{n-1} - a_{n-1}) = \frac{1}{2}\left(\frac{1}{2}\right)^{n-1}(b_0 - a_0) = \left(\frac{1}{2}\right)^n (b_0 - a_0)$$

which proves the estimate. To obtain the bound, we observe that

$$\left(\frac{1}{2}\right)^n(b-a) \leq \epsilon$$

Taking log on both sides, we get the desired bound. Burden and Faires [2010]

ADVANTAGES OF BISECTION METHOD

1. Convergence is assured: If a root is present in the interval, convergence will always occur.
2. Simplicity: Easy to comprehend and put into practice.
3. No Derivative Required: Suitable for functions that are not differentiable.
4. Robustness: Low function-related presumptions.
5. Predictable Convergence Rate: There is a linear convergence rate.
6. Steady and Dependable: Less likely to deviate.

DISADVANTAGES OF BISECTION METHOD

1. Slow Convergence: Linear convergence rate; high precision requires more iterations.
2. Requires Bracketing Interval: The function's initial sign change must occur within this interval.
3. Ineffective for Complex Roots: Mostly for actual roots.
4. Limited by Function Behaviour: Could overlook several roots that are close together.
5. Inadequate for Discontinuous Functions: Only works on continuous functions.
6. No Shape Consideration: Does not adapt to function's shape.

3.2 False Position (Regula Falsi) Method

The method of False Position (also called Regula Falsi) generates approximations in the same manner as the Secant method, but it includes a test to ensure that the root is always bracketed between successive iterations. Although it is not a method we generally recommend, it illustrates how bracketing can be incorporated. Onassis [b]

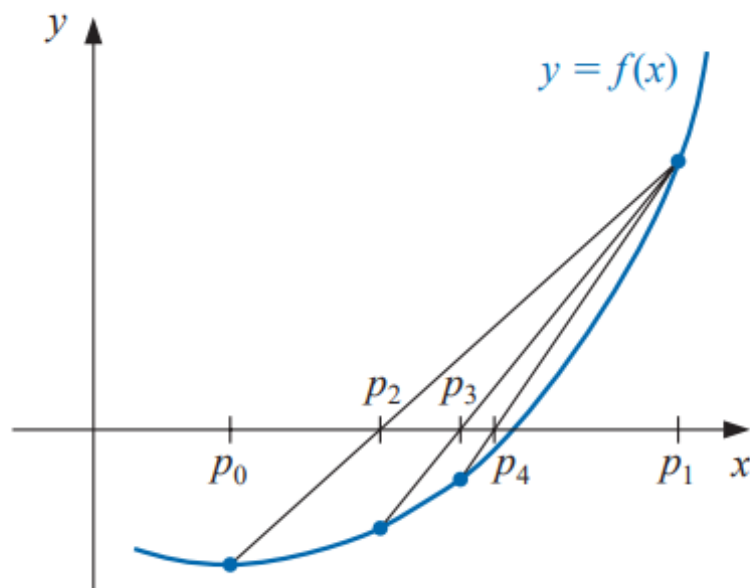


Figure 3.9: A Graph of the False Position Method

PSEUDOCODE FOR FALSE POSITION METHOD

Pseudocode

0. Input : $f(x)=0$, $[a, b]$, tol
Output : $C_n \approx x^*$
1. If $f(a)f(b) > 0$, stop $x^* \notin [a, b]$
2. Set $n=0$, $a_n=a$, $b_n=b$
Repeat until $|b_n - a_n| < \text{tol}$
- 3 $C_n = a_n - \left[\frac{b_n - a_n}{f(b_n) - f(a_n)} \right] f(a_n)$
- 4 If $f(C_n) = 0$, stop : $x^* \approx C_n$
- 5 Set $[a_{n+1}, b_{n+1}] = \begin{cases} [a_n, C_n], & \text{if } f(a_n) \cdot f(C_n) < 0 \\ [C_n, b_n], & \text{if } f(C_n) \cdot f(b_n) < 0 \end{cases}$
- 6 $n = n+1$
End Repeat

Rate of Convergence

Given

$$C_n = a_n - \left[\frac{b_n - a_n}{f(b_n) - f(a_n)} \right] f(a_n)$$

let

$$x_{n+1} = C_n, \quad x_n = b_n \quad \& \quad x_{n-1} = a_n$$

then

$$x_{n+1} = x_{n-1} - \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right] f(x_{n-1})$$

Next let

$$x_n = \alpha + \varepsilon_n, \quad x_{n-1} = \alpha + \varepsilon_{n-1} \quad \& \quad x_{n+1} = \alpha + \varepsilon_{n+1}$$

then

$$\alpha + \varepsilon_{n+1} = \alpha + \varepsilon_{n-1} - \left[\frac{\varepsilon_n - \varepsilon_{n-1}}{f(\alpha + \varepsilon_n) - f(\alpha + \varepsilon_{n-1})} \right] f(\alpha + \varepsilon_{n-1})$$

$$\varepsilon_{n+1} = \varepsilon_{n-1} - \left[\frac{\varepsilon_n - \varepsilon_{n-1}}{f(\alpha + \varepsilon_n) - f(\alpha + \varepsilon_{n-1})} \right] f(\alpha + \varepsilon_{n-1})$$

Using Taylor series expansion (TSE)

$$\begin{aligned} f(\alpha + \varepsilon_{n-1}) &= f(\alpha) + \varepsilon_{n-1} f'(\alpha) + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha) + \dots \\ &= \varepsilon_{n-1} f'(\alpha) + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha) + \dots \end{aligned}$$

$$\begin{aligned} f(\alpha + \varepsilon_n) &= f(\alpha) + \varepsilon_n f'(\alpha) + \frac{1}{2!} \varepsilon_n^2 f''(\alpha) + \dots \\ &= \varepsilon_n f'(\alpha) + \frac{1}{2!} \varepsilon_n^2 f''(\alpha) + \dots \end{aligned}$$

$$\begin{aligned} f(\alpha + \varepsilon_{n+1}) &= f(\alpha) + \varepsilon_{n+1} f'(\alpha) + \frac{1}{2!} \varepsilon_{n+1}^2 f''(\alpha) + \dots \\ &= \varepsilon_{n+1} f'(\alpha) + \frac{1}{2!} \varepsilon_{n+1}^2 f''(\alpha) + \dots \end{aligned}$$

Thus

$$f(\alpha + \varepsilon_n) - f(\alpha + \varepsilon_{n-1}) = (\varepsilon_n - \varepsilon_{n-1}) f'(\alpha) + \frac{1}{2!} (\varepsilon_n^2 - \varepsilon_{n-1}^2) f''(\alpha) + \dots$$

$$= (\varepsilon_n - \varepsilon_{n-1}) f'(\alpha) + \frac{1}{2!} (\varepsilon_n - \varepsilon_{n-1})(\varepsilon_n + \varepsilon_{n-1}) f''(\alpha) + \dots$$

$$\approx (\varepsilon_n - \varepsilon_{n-1}) \left[f'(\alpha) + \frac{1}{2!} (\varepsilon_n + \varepsilon_{n-1}) f''(\alpha) \right]$$

Substing the TSEs in

$$\varepsilon_{n+1} = \varepsilon_{n-1} - \left[\frac{\varepsilon_n - \varepsilon_{n-1}}{f(\alpha + \varepsilon_n) - f(\alpha + \varepsilon_{n-1})} \right] f(\alpha + \varepsilon_{n-1})$$

we obtain

$$\varepsilon_{n+1} = \varepsilon_{n-1} - \frac{(\cancel{\varepsilon_n} - \varepsilon_{n-1})(\varepsilon_{n-1} f'(\alpha) + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha) + \dots)}{(\cancel{\varepsilon_n} - \varepsilon_{n-1})[f'(\alpha) + \frac{1}{2!}(\varepsilon_n + \varepsilon_{n-1})f''(\alpha)]}$$

$$= \varepsilon_{n-1} - \frac{\cancel{f'(\alpha)}(\varepsilon_{n-1} + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha)/f'(\alpha) + \dots)}{\cancel{f'(\alpha)}[1 + \frac{1}{2!}(\varepsilon_n + \varepsilon_{n-1})f''(\alpha)/f'(\alpha)]}$$

$$= \varepsilon_{n-1} - \frac{(\varepsilon_{n-1} + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha)/f'(\alpha) + \dots)}{[1 + \frac{1}{2!}(\varepsilon_n + \varepsilon_{n-1})f''(\alpha)/f'(\alpha)]}$$

$$= \varepsilon_{n-1} - (\varepsilon_{n-1} + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha)/f'(\alpha) + \dots) \underbrace{[1 + \frac{1}{2!}(\varepsilon_n + \varepsilon_{n-1})f''(\alpha)/f'(\alpha)]^{-1}}_{\text{Binomial expansion}}$$

$$= \varepsilon_{n-1} - (\varepsilon_{n-1} + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha)/f'(\alpha) + \dots)[1 - \frac{1}{2!}(\varepsilon_n + \varepsilon_{n-1})f''(\alpha)/f'(\alpha)]$$

$$= \varepsilon_{n-1} - (\varepsilon_{n-1} + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha)/f'(\alpha) + \dots)$$

$$\begin{aligned}
& + \frac{1}{2!}(\varepsilon_{n-1} + \frac{1}{2!} \varepsilon_{n-1}^2 f''(\alpha)/f'(\alpha) + \dots)(\varepsilon_n + \varepsilon_{n-1})f''(\alpha)/f'(\alpha) \\
& \stackrel{\text{Expanding and collecting terms}}{=} \left(-\frac{1}{2!} \varepsilon_{n-1}^2 + \frac{1}{2!} \varepsilon_{n-1}^2 + \frac{1}{2!} \varepsilon_{n-1} \varepsilon_n\right) f''(\alpha)/f'(\alpha) \\
& \quad + \frac{1}{4} \varepsilon_{n-1}^2 (\varepsilon_n + \varepsilon_{n-1}) \left[f''(\alpha)/f'(\alpha)\right]^2 + \dots \\
& = \frac{1}{2!} \varepsilon_{n-1} \varepsilon_n f''(\alpha)/f'(\alpha) + O(\varepsilon_{n-1}^3 + \varepsilon_n \varepsilon_{n-1}^2)
\end{aligned}$$

Thus

$$\begin{aligned}
\varepsilon_{n+1} & \approx \frac{1}{2} \varepsilon_{n-1} \varepsilon_n f''(\alpha)/f'(\alpha) \\
& = C \varepsilon_{n-1} \varepsilon_n
\end{aligned}$$

N.B: In the false position method, one end of the interval remains fixed throughout the iterative process.

ADVANTAGES OF FALSE POSITION METHOD

1. Simplicity and easy to implement.
2. Always converges if f is continuous and initial guesses bracket a root

DISADVANTAGES OF FALSE POSITION METHOD

1. May converge poorly if the function's slope is shallow near the root.
2. Slower convergence compared to methods like the Newton- Raphson

3.3 Newton-Raphson's Method

Newton's (or the Newton-Raphson) method is one of the most powerful and well-known numerical methods for solving a root-finding problem. Suppose that $f \in C^2[a, b]$. Let $p_0 \in [a, b]$ be an approximation to p such that $f(p_0) = 0$ and $|p - p_0|$ is "small." Consider the first Taylor polynomial for $f(x)$ expanded about p_0 and evaluated at $x = p$. Burden and Faires [2010]

$$f(p) = f(p_0) + (p - p_0) f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p))$$

where $\xi(p)$ lies between p and p_0 . Since $f(p) = 0$, this equation gives

$$0 = f(p_0) + (p - p_0) f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p))$$

Newton's method is derived by assuming that since $|p - p_0|$ is small, the term involving $(p - p_0)^2$ is much smaller, so

$$0 \approx f(p_0) + (p - p_0) f'(p_0)$$

Solving for p gives

$$p \approx f(p_0) - \frac{f(p_0)}{f'(p_0)} \equiv p_1$$

This sets the stage for Newton's method, which starts with an initial approximation p_0 and generates the sequence $\{p_n\}_{n=0}^{\infty}$, by

Figure 3.10 illustrates how the approximations are obtained using successive tangents. Starting with the initial approximation p_0 , the approximation p_1 is the x-intercept of the tangent line to the graph of f at $(p_0, f(p_0))$. The approximation p_2 is the x-intercept of the tangent line to the graph of f at $(p_1, f(p_1))$ and so on.

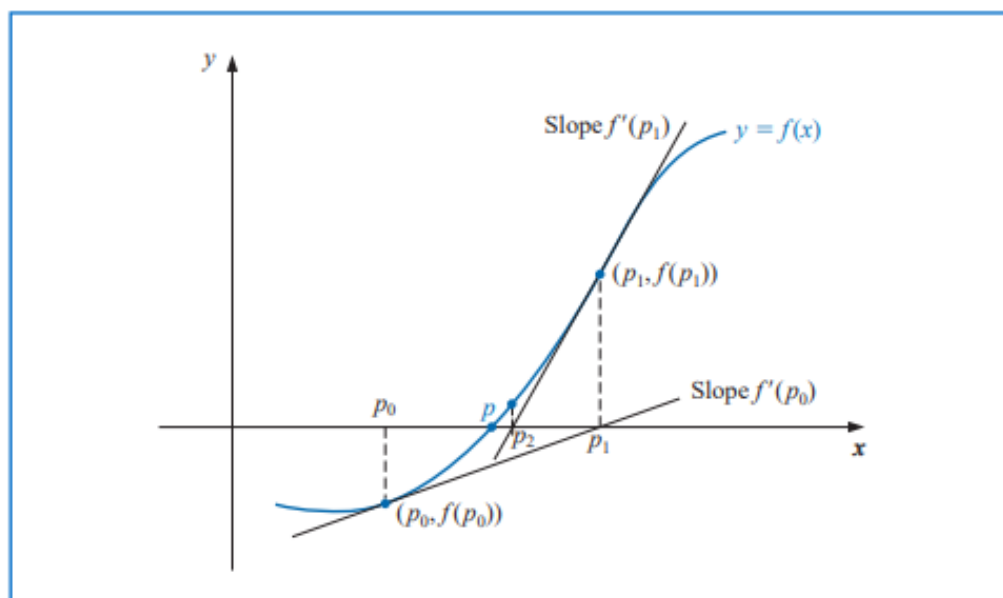


Figure 3.10: A Graph of Newton Raphson's Method

ALGORITHM FOR NEWTON-RAPHSONS METHOD

To find a solution to $f(x) = 0$ given an initial approximation p_0 :

Input: Initial approximation p_0 , tolerance TOL, maximum number of iterations N_0 .

Output: Approximate solution p or message of failure.

1. Set $i = 1$.
2. While $i \leq N_0$, do Steps 3–6.
3. Set $p = p_0 - \frac{f(p_0)}{f'(p_0)}$. (Compute p_i .)
4. If $|p - p_0| < \text{TOL}$, then
5. OUTPUT (p); (The procedure was successful.)
 STOP.
6. Set $i = i + 1$.
7. Set $p_0 = p$. (Update p_0 .)
8. OUTPUT ('The method failed after N_0 iterations, $N_0 =$ ', N_0);
 (The procedure was unsuccessful.)
 STOP.

Stopping Techniques:

The stopping criteria for Newton's method are similar to those for the Bisection method:

1. Select a tolerance $\epsilon > 0$ and construct $p_1, \dots, p_N$ until
 $|p_N - p_{N-1}| < \epsilon$
or
 $\frac{|p_N - p_{N-1}|}{|p_N|} < \epsilon, \quad p_N \neq 0$
or
 $|f(p_N)| < \epsilon$. Burden and Faires [2010]

ADVANTAGES OF NEWTON-RAPHSON'S METHOD

1. Functions near the root with continuous second derivatives exhibit rapid convergence.
2. Generally calls for fewer iterations than other techniques for locating roots, such as the bisection approach.
3. Fit for resolving transcendental and algebraic problems alike.
4. Iterative application is possible to improve solutions with ever-higher accuracy.

DISADVANTAGES OF NEWTON-RAPHSON'S METHOD

1. Demands that the derivative $f'(x)$ be continuous and nonzero close to the root, which could be challenging or costly to compute.

2. The sensitivity of the first guess, p_0 , can cause divergence or convergence to the incorrect root.
3. For functions with complex behaviour close to the root, it may not converge or oscillate.
4. The method does not guarantee convergence for all functions, especially when the derivative changes sign or is close to zero.

3.4 Secant Method

Newton's method is an extremely powerful technique, but it has a major weakness. The need to know the value of the derivative of f at each approximation. Frequently, $f'(x)$ is far more difficult and needs more arithmetic operations to calculate than $f(x)$. To circumvent the problem of the derivative evaluation in Newton's method, we introduce a slight variation. By definition,

$$f'(p_{n-1}) = \lim_{x \rightarrow p_{n-1}} \frac{f(x) - f(p_{n-1})}{x - p_{n-1}}.$$

If p_{n-2} is close to p_{n-1} , then

$$f'(p_{n-1}) \approx \frac{f(p_{n-2}) - f(p_{n-1})}{p_{n-2} - p_{n-1}} = \frac{f(p_{n-1}) - f(p_{n-2})}{p_{n-1} - p_{n-2}}.$$

Using this approximation for $f'(p_{n-1})$ in Newton's formula gives

$$p_n = p_{n-1} - f(p_{n-1}) \frac{p_{n-1} - p_{n-2}}{f(p_{n-1}) - f(p_{n-2})}. \quad (2.12)$$

The word secant is derived from the Latin word *secan*, which means to cut. The secant method uses a secant line, a line joining two points that cut the curve, to approximate a root. Burden and Faires [2010]

This technique is called the Secant method. Starting with the two initial approximations p_0 and p_1 , the approximation p_2 is the x-intercept of the line joining $(p_0, f(p_0))$ and $(p_1, f(p_1))$. The approximation p_3 is the x-intercept of the line joining $(p_1, f(p_1))$ and $(p_2, f(p_2))$, and so on. Note that only one function evaluation is needed per step for the Secant method after p_2 has been determined. In contrast, each step of Newton's method requires an evaluation of both the function and its derivative.

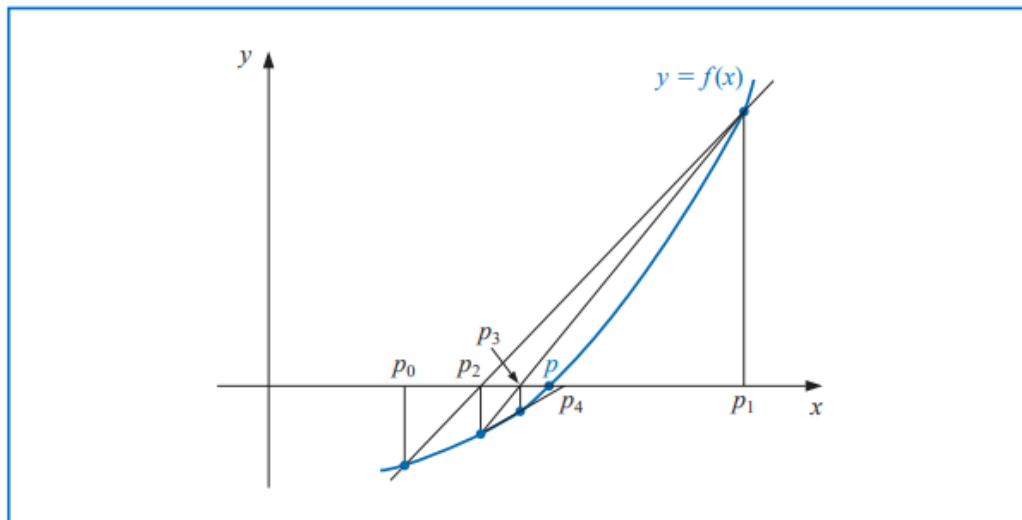


Figure 3.11: A Graph of the Secant Method

ALGORITHM FOR SECANT METHOD

According to Burden and Faires [2010] to find a solution to $f(x) = 0$ given initial approximations p_0 and p_1 :

INPUT initial approximations(p_0, p_1); tolerance (TOL); maximum number of iterations (N_0).

OUTPUT approximate solution p or message of failure.

1. Set $i = 2$;

$$q_0 = f(p_0);$$

$$q_1 = f(p_1).$$

2. While $i \leq N_0$ do Steps 3–6.

3. Set

$$p = p_1 - q_1 \frac{p_1 - p_0}{q_1 - q_0}.$$

(Compute p_i .)

4. If $|p - p_1| < \text{TOL}$ then

(a) OUTPUT p ; (The procedure was successful.)

(b) STOP.

5. Set $i = i + 1$.

6. Update p_0, q_0, p_1, q_1 :

$$p_0 = p_1;$$

$$q_0 = q_1;$$

$$p_1 = p;$$

$$q_1 = f(p).$$

7. OUTPUT ‘The method failed after N_0 iterations, $N_0 =', N_0$; (The procedure was unsuccessful.)
8. STOP.

ADVANTAGES OF SECANT METHOD

1. It converges faster than a linear rate, so that it is more rapidly convergent than the bisection method.
2. It does not require use of the derivative of the function, something that is not available in a number of applications.
3. It requires only one function evaluation per iteration, as compared with Newton’s method which requires two.

DISADVANTAGES OF SECANT METHOD:

1. It may not converge.
2. There is no guaranteed error bound for the computed iterates.
3. It is likely to have difficulty if $f'(\alpha) = 0$. This means the x-axis is tangent to the graph of $y = f(x)$ at $x = \alpha$.

3.5 Fixed Point Method

The fixed point iteration method in numerical analysis is used to find an approximate solution to algebraic and transcendental equations. Sometimes, it becomes very tedious to find solutions to cubic, bi-quadratic and transcendental equations; then, we can apply specific numerical methods to find the solution; one among those methods is the fixed point iteration method. Epperson [2007].

Suppose we have an equation $f(x) = 0$, for which we have to find the solution. The equation can be expressed as $x = g(x)$. Choose $g(x)$ such that $|g'(x)| < 1$ at $x = x_0$, where x_0 is some initial guess called fixed point iterative scheme. Then the iterative method is applied by successive approximations given by $x_n = g(x_{n-1})$, that is, $x_1 = g(x_0)$, $x_2 = g(x_1)$, and so on.

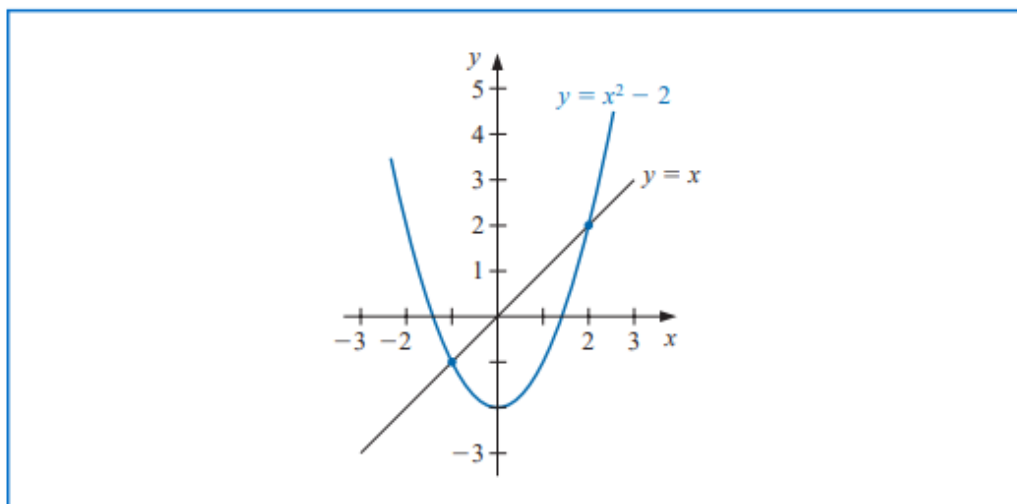


Figure 3.12: A Graph of the Secant Method

STEPS OF FIXED POINT ITERATION

1. **Transform the Equation:** Express the equation $f(x) = 0$ in the form $x = g(x)$.
2. **Initial Guess:** Choose an initial guess x_0 .
3. **Iterative Process:** Apply the iteration $x_n = g(x_{n-1})$ until convergence.

CONVERGENCE CRITERIA

Contraction Mapping: A function g is a contraction mapping on a closed interval $[a, b]$ if there exists a constant $0 \leq k < 1$ such that for all $x, y \in [a, b]$:

$$|g(x) - g(y)| \leq k|x - y|$$

According to the Banach Fixed Point Theorem, a contraction mapping on a closed interval has a unique fixed point, and the iteration will converge to this fixed point for any initial guess within the interval.

ADVANTAGES OF FIXED POINT METHOD

1. Easy to understand and implement, involving repeated application of a function $g(x)$.
2. Each iteration requires only one function evaluation and simple arithmetic operations.
3. Can be extended to solve systems of equations using vector-valued functions and vector iterations.

4. Solves a variety of nonlinear equations.

DISADVANTAGES OF FIXED POINT METHOD

1. Convergence is typically local, not global.
2. Can be slow, especially if the function is not well-behaved.
3. May not converge if the function $g(x)$ is not chosen appropriately.
4. Less effective for highly nonlinear problems.

3.6 Gaussian Elimination Method

The systematic elimination of variables to change a system of linear equations into an equivalent system in echelon form from which we can read the solution is called Gaussian elimination in honor of Carl Friedrich Gauss (1777-1855), one of the most brilliant mathematicians of all time. The key to Gaussian elimination (which can be done efficiently on computers) is the idea of elementary operation, the replacement of one equation in a system by another in a way that leaves the solution unchanged. Each of the following operations gives an equivalent system, that has the same solution set. Burden and Faires [2010]

STEPS IN GAUSSIAN ELIMINATION

The process consists of two main steps: forward elimination and back substitution.

FORWARD ELIMINATION

1. Transform the given system of equations into an upper triangular matrix form.
2. For a system $A\mathbf{x} = \mathbf{b}$, where A is an $n \times n$ matrix and $\mathbf{b}$ is a vector, perform row operations to introduce zeros below the main diagonal.
3. The operations include:
 - (a) **Swapping rows:** $R_i \leftrightarrow R_j$
 - (b) **Multiplying a row by a nonzero scalar:** $R_i \rightarrow kR_i$ where $k \neq 0$
 - (c) **Adding a multiple of one row to another:** $R_j \rightarrow R_j + kR_i$. Epperson [2007]

BACK SUBSTITUTION

1. Once the system is in upper triangular form, solve for the variables starting from the last equation and moving upwards.

2. For a matrix in the form:

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ 0 & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} & b_n \end{array} \right)$$

solve for x_n from the last row, then substitute x_n into the previous equations to find $x_{n-1}, x_{n-2}, \dots, x_1$. Epperson [2007]

ADVANTAGES OF GAUSSIAN ELIMINATION METHOD

1. Provides a clear, step-by-step procedure to solve linear systems.
2. Can be applied to any square matrix, regardless of its properties (as long as it is non-singular).
3. Forms the basis for more advanced techniques like LU decomposition.
4. Helps in checking the consistency of a system and finding the rank of a matrix.

DISADVANTAGES OF GAUSSIAN ELIMINATION METHOD

1. For large systems, the method can be computationally expensive (approximately $O(n^3)$ operations for an $n \times n$ system).
2. Without partial pivoting, the method can suffer from numerical instability.
3. For sparse systems, specialized methods like iterative solvers are often more efficient.
4. Round-off errors can accumulate during the elimination process, especially for ill-conditioned matrices.

3.7 Trapezoidal Rule

The Trapezoidal rule is a numerical method used to approximate the definite integral of a function. It is particularly useful when an analytical solution to the integral is difficult or impossible to obtain. The method works by approximating the area under a curve by dividing it into trapezoids and summing their areas. To derive the Trapezoidal rule for approximating $\int_a^b f(x) dx$, let $x_0 = a$, $x_1 = b$, and $h = b - a$. We use the linear Lagrange polynomial: Burden and Faires [2010]

$$P_1(x) = \frac{(x - x_1)}{(x_0 - x_1)}f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)}f(x_1).$$

Then,

$$\int_a^b f(x) dx = \int_{x_0}^{x_1} \left(\frac{(x-x_1)}{(x_0-x_1)} f(x_0) + \frac{(x-x_0)}{(x_1-x_0)} f(x_1) \right) dx + \frac{1}{2} \int_{x_0}^{x_1} f(\xi(x))(x-x_0)(x-x_1) dx.$$

The product $(x-x_0)(x-x_1)$ does not change sign on $[x_0, x_1]$, so the Weighted Mean Value Theorem for Integrals can be applied to the error term to give, for some ξ in (x_0, x_1) ,

$$\begin{aligned} \int_{x_0}^{x_1} f(\xi(x))(x-x_0)(x-x_1) dx &= f(\xi) \int_{x_0}^{x_1} (x-x_0)(x-x_1) dx \\ &= f(\xi) \left[\frac{x^3}{3} - \frac{(x_1+x_0)}{2} x^2 + x_0 x_1 x \right]_{x_0}^{x_1} \\ &= -\frac{h^3}{6} f(\xi). \end{aligned}$$

$$\begin{aligned} \int_a^b f(x) dx &= \left[\frac{(x-x_1)^2}{2(x_0-x_1)} f(x_0) + \frac{(x-x_0)^2}{2(x_1-x_0)} f(x_1) \right]_{x_0}^{x_1} - \frac{h^3}{12} f(\xi) \\ &= \frac{(x_1-x_0)}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f(\xi). \end{aligned}$$

Using the notation $h = x_1 - x_0$ gives the following rule:

Trapezoidal Rule:

$$\int_a^b f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f(\xi).$$

This is called the Trapezoidal rule because when f is a function with positive values, $\int_a^b f(x) dx$ is approximated by the area in a trapezoid.

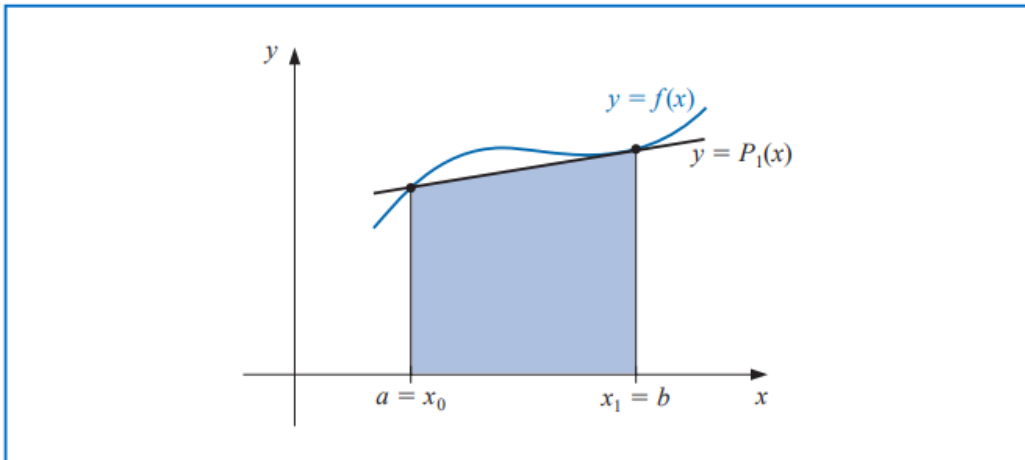


Figure 3.13: A Graph of the Trapezoidal Rule

The error term for the Trapezoidal rule involves f'' , so the rule gives the exact result when applied to any function whose second derivative is identically zero, that is, any polynomial of degree one or less. Gautschi [2001]

ADVANTAGES OF THE TRAPEZOIDAL RULE

1. The Trapezoidal rule is straightforward to understand and implement, making it a popular choice for numerical integration.
1. It can be applied to a wide range of continuous functions, making it versatile for different types of problems.
1. For functions that are reasonably smooth, the Trapezoidal rule can provide good approximations with relatively low computational effort.

DISADVANTAGES OF THE TRAPEZOIDAL RULE

1. Less accurate for functions with high curvature or rapid changes in the interval of integration.
1. Since it approximates the function by linear segments, it may not be accurate for functions that are highly nonlinear within each subinterval.
1. The error term involves the second derivative of the function. For functions where the second derivative is large, the Trapezoidal rule may yield poor results.

3.8 Simpson's 1/3 Rule

Simpson's 1/3 Rule is a numerical method used for approximating the definite integral of a function. It is based on approximating the function $f(x)$ over each pair of subintervals using quadratic polynomials. This rule provides a more accurate approximation compared to simpler methods like the Trapezoidal rule. Simpson's rule results from integrating over $[a, b]$ the second Lagrange polynomial with equally-spaced nodes $x_0 = a$, $x_2 = b$, and $x_1 = a + \frac{b-a}{2}$, where $h = \frac{b-a}{2}$:

$$\int_a^b f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] + O(h^5)$$

This formula provides an approximation to the integral of a function $f(x)$ over the interval $[a, b]$, where h is the step size between the nodes.

Therefore,

$$\int_a^b f(x) dx = \frac{x_2 - x_0}{x_2 - x_0} \left[\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2) \right]$$

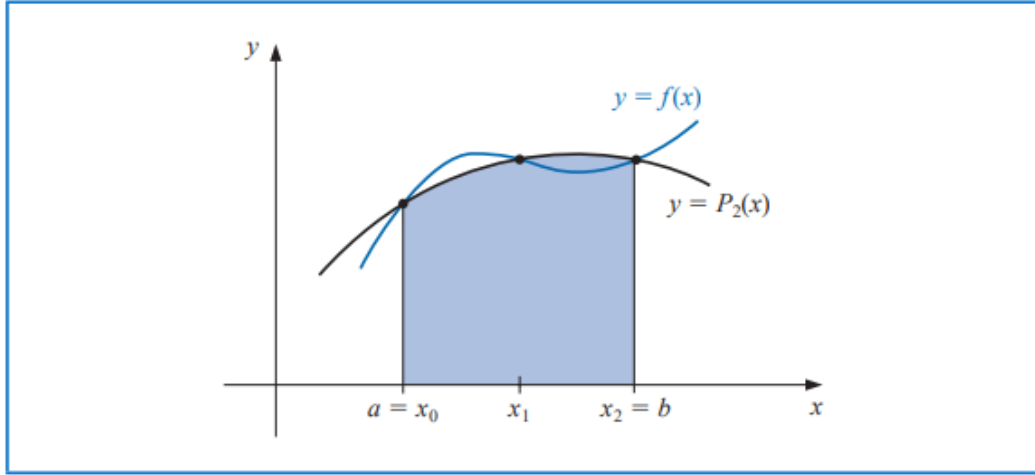


Figure 3.14: A Graph of the Simpson's 1/3 Rule

$$+ \frac{x_2 - x_0}{x_2 - x_0} \frac{(x - x_0)(x - x_1)(x - x_2)}{6} f^{(3)}(\xi(x)) dx.$$

Deriving Simpson's rule in this manner, however, provides only an $O(h^4)$ error term involving $f^{(3)}$. By approaching the problem in another way, a higher-order term involving $f^{(4)}$ can be derived. Kincaid and Cheney [2009]

To illustrate this alternative method, suppose that f is expanded in the third Taylor polynomial about x_1 . Then for each x in $[x_0, x_2]$, a number $\xi(x)$ in (x_0, x_2) exists with

$$f(x) = f(x_1) + f'(x_1)(x - x_1) + \frac{f''(x_1)}{2}(x - x_1)^2 + \frac{f^{(3)}(\xi(x))}{6}(x - x_1)^3 + \frac{f^{(4)}(\xi(x))}{24}(x - x_1)^4,$$

and

$$\begin{aligned} \int_a^b f(x) dx &= \left[f(x_1)(x - x_1) + \frac{f'(x_1)}{2}(x - x_1)^2 + \frac{f''(x_1)}{6}(x - x_1)^3 + \frac{f^{(4)}(\xi(x))}{24}(x - x_1)^4 \right]_{x_0}^{x_2} \\ &+ \frac{1}{24} \int_{x_0}^{x_2} f^{(4)}(\xi(x)) \frac{(x - x_1)^4}{dx} \end{aligned}$$

Because $(x - x_1)^4$ is never negative on $[x_0, x_2]$, the Weighted Mean Value Theorem for the implies that

$$\frac{1}{24} \int_{x_0}^{x_2} f^{(4)}(\xi(x))(x - x_1)^4 dx = \frac{f^{(4)}(\xi_1)}{24} \int_{x_0}^{x_2} (x - x_1)^4 dx = \frac{f^{(4)}(\xi_1)}{120} (x - x_1)^5 \Big|_{x_0}^{x_2},$$

for some number ξ_1 in (x_0, x_2) . However, $h = x_2 - x_1 = x_1 - x_0$, so

$$(x_2 - x_1)^2 - (x_0 - x_1)^2 = (x_2 - x_1)^4 - (x_0 - x_1)^4 = 0,$$

whereas

$$(x_2 - x_1)^3 - (x_0 - x_1)^3 = 2h^3 \quad \text{and} \quad (x_2 - x_1)^5 - (x_0 - x_1)^5 = 2h^5.$$

$$\int_{x_0}^{x_2} f(x) dx = 2hf(x_1) + \frac{h^3}{3}f'(x_1) + \frac{f^{(4)}(\xi_1)}{60}h^5.$$

If we now replace $f(x_1)$ by the approximation given in Eq. (4.9) of Section 4.1, we have

$$\begin{aligned}\int_{x_0}^{x_2} f(x) dx &= 2hf(x_1) + \frac{h^3}{3} \frac{1}{h^2} [f(x_0) - 2f(x_1) + f(x_2)] - \frac{h^2}{12} f^{(4)}(\xi_2) + \frac{f^{(4)}(\xi_1)}{60} \frac{h^5}{h^4} \\ &= \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{12} \left[\frac{1}{3} f^{(4)}(\xi_2) - \frac{1}{5} f^{(4)}(\xi_1) \right].\end{aligned}$$

It can be shown by alternative methods that the values ξ_1 and ξ_2 in this expression can be replaced by a common value ξ in (x_0, x_2) . This gives Simpson's rule.

Simpson's Rule:

$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90} f^{(4)}(\xi).$$

The error term in Simpson's rule involves the fourth derivative of f , so it gives exact results when applied to any polynomial of degree three or less.

ADVANTAGES OF THE SIMPSON'S $\frac{1}{3}$ RULE

1. Simpson's $1/3$ Rule provides more accurate results compared to the Trapezoidal rule, especially for smooth functions, due to its use of quadratic polynomials.
2. It gives exact results for polynomials of degree three or less, which is advantageous in many numerical integration applications.
3. It is less sensitive to the step size h compared to other numerical methods, making it more robust in practical applications.

DISADVANTAGES OF THE SIMPSON'S $\frac{1}{3}$ RULE

1. Implementing Simpson's $1/3$ Rule for intervals with unequal widths can be complex and may require additional adjustments.
2. It may not be as straightforward to apply when the number of intervals n is odd, which can complicate the integration process.
3. Compared to simpler methods like the Trapezoidal rule, Simpson's $1/3$ Rule can require more computational effort due to its quadratic polynomial approximation.

3.9 LU Decomposition

The most well-known and the first matrix decomposition we should be aware of is the LU decomposition, where L stands for lower triangular and U stands for upper triangular. It factors a matrix into a lower triangular matrix and an upper triangular matrix. The LU decomposition is closely associated with Gaussian elimination, and both methods are

commonly employed for solving linear systems, computing determinants, and inverting matrices.

LU Decomposition with Permutation

Every nonsingular $n \times n$ square matrix A can be factored as

$$A = PLU,$$

where P is a permutation matrix, L is a unit lower triangular matrix (i.e., a lower triangular matrix with all 1's on the diagonal), and U is a nonsingular upper triangular matrix.

One of the main applications of LU decomposition is in solving linear systems. Given a system of linear equations, $Ax = b$, where A is a square matrix and b is a target vector, the LU decomposition can be used to find x efficiently. First, the matrix A is factored into P , L , and U matrices, and then the system of equations is rewritten as $Pw = b$, $Lv = w$, and $Ux = v$. The system can then be solved using forward and backward substitutions. The LU decomposition is particularly useful when solving systems of equations that have the same coefficient matrix but different right-hand sides, i.e., solving the n linear equations $Ax = b_1$, $Ax = b_2$, $\dots$, $Ax = b_n$ simultaneously.

LU decomposition is also used in matrix inversion, where the inverse of a matrix A is calculated as $A^{-1} = U^{-1}L^{-1}P^T$. However, this method is not always the most efficient way to compute the inverse of a matrix, and other methods such as the Gauss-Jordan method or the QR decomposition may be more suitable for some applications.

Another application of LU decomposition is in computing the determinant of a matrix. The determinant of a matrix can be calculated as the product of the diagonal elements of the U matrix, multiplied by the product of the diagonal elements of the L matrix and a constant (-1) raised to the power of the number of row exchanges performed during the LU decomposition. Epperson [2007]

LU Decomposition

To factor the $n \times n$ matrix $A = [a_{ij}]$ into the product of the lower-triangular matrix $L = [l_{ij}]$ and the upper-triangular matrix $U = [u_{ij}]$; that is, $A = LU$, where the main diagonal of either L or U consists of all ones:

INPUT

1. Dimension n ;
2. The entries a_{ij} , $1 \leq i, j \leq n$ of A ;
3. The diagonal $l_{11} = \dots = l_{nn} = 1$ of L or the diagonal $u_{11} = \dots = u_{nn} = 1$ of U .

OUTPUT

1. The entries l_{ij} , $1 \leq j \leq i$, $1 \leq i \leq n$ of L ;
2. The entries u_{ij} , $i \leq j \leq n$, $1 \leq i \leq n$ of U .

Algorithm

1. **Step 1:** Select l_{11} and u_{11} satisfying $l_{11}u_{11} = a_{11}$. If $l_{11}u_{11} = 0$ then OUTPUT ('Factorization impossible'); STOP.
2. **Step 2:** For $j = 2, \dots, n$:
 - (a) Set $u_{1j} = \frac{a_{1j}}{l_{11}}$; (First row of U)
 - (b) Set $l_{j1} = \frac{a_{j1}}{u_{11}}$; (First column of L)
3. **Step 3:** For $i = 2, \dots, n-1$ do Steps 4 and 5.
4. **Step 4:** Select l_{ii} and u_{ii} satisfying:

$$l_{ii}u_{ii} = a_{ii} - \sum_{k=1}^{i-1} l_{ik}u_{ki}$$

If $l_{ii}u_{ii} = 0$ then OUTPUT ('Factorization impossible'); STOP.

5. **Step 5:** For $j = i+1, \dots, n$:
 - (a) Set $u_{ij} = \frac{1}{l_{ii}} \left(a_{ij} - \sum_{k=1}^{i-1} l_{ik}u_{kj} \right)$; (i-th row of U)
 - (b) Set $l_{ji} = \frac{1}{u_{ii}} \left(a_{ji} - \sum_{k=1}^{i-1} l_{jk}u_{ki} \right)$; (i-th column of L)
6. **Step 6:** Select l_{nn} and u_{nn} satisfying:

$$l_{nn}u_{nn} = a_{nn} - \sum_{k=1}^{n-1} l_{nk}u_{kn}$$

(Note: If $l_{nn}u_{nn} = 0$, then $A = LU$ but A is singular.)

7. **Step 7:** OUTPUT (l_{ij} for $j = 1, \dots, i$ and $i = 1, \dots, n$); OUTPUT (u_{ij} for $j = i, \dots, n$ and $i = 1, \dots, n$); STOP.

Once the matrix factorization is complete, the solution to a linear system of the form $Ax = b$ using LU is found by first letting $y = Ux$ and solving $Ly = b$ for y . Since L is lower triangular, we have:

$$y_1 = \frac{b_1}{l_{11}}$$

and, for each $i = 2, 3, \dots, n$:

$$y_i = \frac{1}{l_{ii}} \left(b_i - \sum_{j=1}^{i-1} l_{ij}y_j \right)$$

After y is found by this forward-substitution process, the upper-triangular system $Ux = y$ is solved for x by backward substitution using the equations:

$$x_n = \frac{y_n}{u_{nn}}$$

and for $i = n-1, n-2, \dots, 1$:

$$x_i = \frac{1}{u_{ii}} \left(y_i - \sum_{j=i+1}^n u_{ij}x_j \right)$$

Advantages of the LU Decomposition Method

1. **Efficiency:** Solves multiple linear systems quickly once the matrix is decomposed.
2. **Stability:** More stable than direct methods like Gaussian elimination when using pivoting.
3. **Reusability:** The decomposition can be reused for solving different systems with the same matrix.

Disadvantages of the LU Decomposition Method

1. **Pivoting Needed:** Numerical stability requires extra steps for pivoting, adding complexity.
2. **Memory Usage:** Requires extra memory to store both the lower and upper triangular matrices.
3. **Not Always Applicable:** May not work for singular or nearly singular matrices.

JACOBI METHOD

The Jacobi method is a classical iterative algorithm used to solve systems of linear equations of the form

$$A\mathbf{x} = \mathbf{b},$$

where A is a square matrix, and $\mathbf{x}$ and $\mathbf{b}$ are vectors. Named after the German mathematician Carl Gustav Jacob Jacobi, this method is particularly useful for large systems where direct methods like Gaussian elimination become computationally expensive. The Jacobi method can be written in the form

$$\mathbf{x}^{(k)} = T\mathbf{x}^{(k-1)} + \mathbf{c}$$

by splitting A into its diagonal and off-diagonal parts. To see this, let D be the diagonal matrix whose diagonal entries are those of A , $-L$ be the strictly lower-triangular part of A , and $-U$ be the strictly upper-triangular part of A . With this notation,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

is split into

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix} - \begin{bmatrix} 0 & 0 & \cdots & 0 \\ -a_{21} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ -a_{n1} & -a_{n2} & \cdots & 0 \end{bmatrix} - \begin{bmatrix} 0 & -a_{12} & \cdots & -a_{1n} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & 0 \end{bmatrix} = D - L - U.$$

The equation $A\mathbf{x} = \mathbf{b}$, or $(D - L - U)\mathbf{x} = \mathbf{b}$, is then transformed into

$$D\mathbf{x} = (L + U)\mathbf{x} + \mathbf{b},$$

and, if D^{-1} exists, that is, if $a_{ii} \neq 0$ for each i , then

$$\mathbf{x} = D^{-1}(L + U)\mathbf{x} + D^{-1}\mathbf{b}.$$

This results in the matrix form of the Jacobi iterative technique:

$$\mathbf{x}^{(k)} = D^{-1}(L + U)\mathbf{x}^{(k-1)} + D^{-1}\mathbf{b}, \quad k = 1, 2, \dots$$

Introducing the notation $T_j = D^{-1}(L + U)$ and $\mathbf{c}_j = D^{-1}\mathbf{b}$ gives the Jacobi technique the form

$$\mathbf{x}^{(k)} = T_j\mathbf{x}^{(k-1)} + \mathbf{c}_j.$$

Gilat and Subramaniam [2008]

Advantages of the Jacobi Method

1. **Simplicity:** The Jacobi method is easy to understand and implement, making it a straightforward choice for solving linear systems iteratively.
2. **Parallelization:** The method allows for parallel computation since the updates to each variable are independent of each other, enabling efficient use of multi-core processors.
3. **Memory Efficiency:** It requires less memory, especially when dealing with sparse matrices, as only the non-zero elements of the matrix need to be stored.

Disadvantages of the Jacobi Method

1. **Slow Convergence:** The Jacobi method often converges more slowly compared to other iterative methods like the Gauss-Seidel method, especially for large or ill-conditioned systems.
2. **Convergence Limitations:** The method may not converge if the matrix A is not diagonally dominant or symmetric positive definite, limiting its applicability to certain types of systems.
3. **Sensitivity to Initial Guess:** The number of iterations required for convergence can be highly sensitive to the initial guess, with poor guesses leading to significantly slower convergence.

Implementation and Results

4.1 Introduction

In this chapter, we will discuss the implementation process of the NumLab web application, which was developed to solve numerical analysis problems efficiently. The chapter will cover the system design, the technologies used, the development process, testing, validation, and user experience. The results of the application's performance and feedback from users will also be presented.

4.2 System Design

4.2.1 Architecture Overview

NumLab is built on a web-based architecture using a combination of front-end and back-end technologies. The front-end handles the user interface, while the back-end is responsible for the computation of numerical algorithms. The application is hosted on a Flask server, with HTML and CSS providing the structure and styling, and JavaScript enhancing interactivity. Python is used to implement the numerical algorithms, and MathJax is integrated to render mathematical equations in the browser.

4.2.2 Technologies Used

1. HTML and CSS: For structuring and styling the web pages.
2. Flask: A lightweight web framework used to handle server-side logic.
3. Python: The primary language for implementing numerical algorithms.
4. MathJax: A JavaScript library that displays mathematical notations in the browser.
5. JavaScript: For enhancing user interactivity and managing client-side logic.
6. NumPy and SciPy: Python libraries used for numerical computations.
7. Git and VS Code: For version control and code editing, respectively.

4.2.3 Module Description

1. User Interface Module: Handles user inputs and displays results.
2. Computation Module: Implements the numerical methods.
3. Visualization Module: Generates graphs and visual representations of results.
4. Error Handling Module: Manages and reports errors or exceptions.

4.3 System Architecture

4.3.1 Overview of NumLab's Architecture

NumLab is designed as a client-server web application where the front-end is responsible for user interaction, while the back-end handles computational tasks. The application follows a modular architecture, ensuring that each component functions independently while being seamlessly integrated.

4.3.2 Front-End Technologies

The front-end of NumLab is developed using HTML5 for structuring the content, CSS3 for styling and ensuring responsiveness, and JavaScript to handle dynamic content. The use of MathJax is crucial for rendering mathematical equations, making it easier for users to input and view complex mathematical expressions.

1. **HTML5:** It forms the backbone of the application's structure, enabling the creation of input forms, results display areas, and other content sections.
2. **CSS3:** CSS3 is employed to style the application, ensuring that it looks visually appealing and is responsive across different devices, including desktops, tablets, and mobile phones.
3. **JavaScript:** JavaScript is used for client-side scripting, enabling dynamic content updates without the need for a full page reload. The framework chosen (e.g., React) simplifies the management of complex user interactions.
4. **MathJax:** This library is integrated to handle the display of mathematical equations, ensuring that users can both input and view expressions in a familiar format.

4.3.3 Back-End Technologies

The back-end is powered by Python, chosen for its extensive libraries that are well-suited for numerical computations. Flask serves as the web framework, providing the necessary infrastructure for handling requests and responses.

1. **Python:** Python is the primary language for implementing the numerical methods due to its simplicity and the powerful libraries it offers for mathematical computations.
2. **Flask:** Flask is selected as the web framework due to its lightweight nature and ease of integration with Python libraries. It manages the routing, handles HTTP requests, and interacts with the front-end.
3. **RESTful API:** A RESTful API is implemented to facilitate communication between the front-end and back-end. The API endpoints are designed to receive inputs from the user, process them using the implemented numerical methods, and return the results.

4.4 Development Environment and Tools

4.4.1 Development Stack

NumLab was developed using a robust and efficient development stack to ensure high-quality code and streamlined workflows.

1. **IDE:** Visual Studio Code was chosen as the IDE for its support for multiple languages (Python, HTML, CSS) and its extensive set of plugins that enhance productivity.
2. **Version Control:** Git is used for version control, with GitHub hosting the repository. This allows for easy tracking of changes, collaborative development, and rollback of code when necessary.
3. **Containerization:** Docker is employed to ensure that the development environment remains consistent across different systems. This prevents issues that arise from differences in development environments.

4.4.2 Libraries and Frameworks

NumLab relies on several key libraries and frameworks to handle numerical computations and web development:

1. **NumPy:** NumPy is used extensively for handling arrays and performing numerical operations efficiently.
2. **SciPy:** SciPy is employed for more advanced scientific computations, particularly in implementing the numerical methods discussed in Chapter 3.
3. **SymPy:** SymPy is used for symbolic mathematics, enabling the manipulation and solving of mathematical expressions symbolically.

4.4.3 Justification for Technology Choices

Each technology used in NumLab was chosen for its suitability in handling the tasks required:

1. **Python:** Python's simplicity and the extensive libraries it offers for numerical analysis made it the ideal choice for this project.
2. **Flask:** Flask was chosen for its lightweight nature and the ease with which it can be integrated with Python and other libraries.
3. **NumPy, SciPy, and SymPy:** These libraries are essential for efficient numerical computation and symbolic mathematics, offering the necessary tools to implement the numerical methods accurately.

4.5 Implementation of Numerical Methods

For each numerical method discussed in Chapter 3 (e.g., Bisection, False Position, Newton-Raphson):

4.5.1 Detailed Explanation of the Implementation Process

Each method was implemented following a systematic approach. The algorithm was first designed and tested in isolation, ensuring its correctness before integrating it into the web application.

4.5.2 Code Snippets or Pseudocode

Code snippets or pseudocode is provided in Figure 4.5.2 to illustrate the implementation of the key algorithms. This includes the core logic for methods like Bisection and Newton-Raphson, showing how inputs are processed and results are computed. Hence we're we're looking at that of bisection

```
from sympy import symbols, lambdify, parse_expr

def bisection_method(equation_str, a, b, tol=1e-6, max_iter=1000):
    iterations = 0
    data = []
    intervals = []

    x = symbols('x')
    equation = parse_expr(equation_str)
    equation = lambdify(x, equation)

    if equation(a) * equation(b) >= 0:
        raise ValueError("Bisection method fails.")

    while (b - a) / 2 > tol and iterations < max_iter:
        c = (a + b) / 2
        root_estimate = (a + b) / 2
        func_value = equation(root_estimate)
        data.append((iterations, root_estimate, func_value))

        # Save the current interval
        intervals.append((a, b))

        if func_value == 0:
            break
        if equation(c) * equation(a) < 0:
            b = root_estimate
        else:
            a = root_estimate
        iterations += 1

    return root_estimate, data, intervals
```

4.5.3 Optimization Techniques

To enhance the performance of NumLab, several optimizations were made:

1. **Algorithmic Optimization:** Reducing the number of iterations required for convergence, improving the accuracy of results.
2. **Performance Considerations:** Ensuring that the application can handle complex calculations without significant delays, optimizing the backend processes to minimize computation time.

4.5.4 Challenges and Solutions

During the implementation, several challenges were encountered, including handling edge cases where numerical methods might fail to converge. These were addressed by refining the algorithms and implementing fallback solutions where necessary.

4.6 User Interface Design

4.6.1 Design Philosophy

The design of NumLab follows a principle of simplicity and intuitiveness, ensuring that users, regardless of their expertise level, can navigate the application with ease.

4.6.2 User Experience Considerations

The user experience was a key consideration, with the interface designed to guide users through the process of inputting problems and viewing results. Clear instructions and visual feedback were implemented to enhance the overall experience.

4.6.3 Wireframes or Mockups

Wireframes and mockups would be provided to illustrate the design process, showing how the layout was planned and the rationale behind the placement of various components.

4.6.4 Responsive Design Implementation

The UI was designed to be responsive, ensuring that it functions well on different devices. Special attention was given to mobile users, ensuring that the application remains usable even on smaller screens.

4.7 Features and Functionality

4.7.1 Detailed Breakdown of NumLab's Features

NumLab offers a range of features designed to assist users in solving numerical problems. These include:

1. **Solution Finder:** Users can enter their equations through a user friendly interface and receive a step-by-step solution, guiding them through the problem-solving process.
2. **Error Handling:** The application includes robust error handling, providing feedback to users when inputs are invalid or when methods fail to converge.
3. **Feedback:** The application also includes a feedback mechanism, allowing users to report issues or suggest improvements, ensuring continuous development and refinement.

4.7.2 Problem Input Mechanisms

The input mechanisms are designed to be intuitive, allowing users to enter problems using standard mathematical notation. The inputs are validated to ensure correctness before being processed.

4.7.3 Solution Visualization

NumLab includes powerful visualization tools, using libraries like Matplotlib to plot graphs and display results in a clear and concise manner.

4.7.4 Error Handling and User Guidance

To enhance user experience, NumLab includes error handling that provides clear feedback to users, helping them correct mistakes or adjust inputs as necessary. Additionally, tooltips and help sections are included to guide users through the application.

4.8 Testing and Validation

4.8.1 Integration Testing

Integration testing was conducted to ensure that the different components of NumLab work together seamlessly. This involved testing the communication between the front-end and back-end, as well as the overall user experience.

4.8.2 User Acceptance Testing

User feedback was gathered through user acceptance testing, ensuring that the final product met the needs and expectations of its intended audience. Feedback was used to refine the application before its final release.

4.8.3 Performance Testing

Performance testing was carried out to assess the speed and efficiency of NumLab. The performance of the application was benchmarked against other similar tools and manual calculations. The goal was to ensure that NumLab could handle complex calculations quickly and accurately without excessive resource consumption.

4.8.4 Validation of Numerical Results

To validate the accuracy of the numerical methods implemented in NumLab, the results were compared against benchmark solutions from established numerical analysis software and manual calculations. This comparison ensured that NumLab provided accurate and reliable results.

4.9 Deployment and Maintenance

4.9.1 Deployment Strategy

NumLab was deployed using a cloud-based service like Heroku or AWS, allowing it to be accessible from anywhere with an internet connection. The deployment process included setting up the necessary environment, configuring the server, and ensuring that the application was optimized for performance.

4.9.2 Server Configuration

The server was configured to handle multiple concurrent users, with load balancing and caching strategies implemented to improve performance.

4.9.3 Monitoring and Logging

To maintain the health and performance of the application, monitoring tools were put in place to track usage patterns, errors, and other metrics. Logs were configured to capture detailed information about the application's operation, aiding in debugging and performance tuning.

4.9.4 Continuous Integration/Continuous Deployment (CI/CD)

A CI/CD pipeline was set up to automate the deployment process, ensuring that updates could be rolled out quickly and reliably. This pipeline includes automated testing and deployment steps, reducing the risk of introducing errors with new releases.

4.9.5 Maintenance and Updates

Regular maintenance is planned to ensure that NumLab remains up-to-date with the latest libraries and security patches. Updates will be released periodically to add new features, improve performance, and address any issues that arise.

4.10 Case Studies and Examples

4.10.1 Example Problem 1: Solving Non-linear Equations

Problem Statement: Solve the equation $f(x) = \cos(x) - x$ using the Bisection method.

Solution: The problem was solved using NumLab, with the Bisection method implemented as described in Chapter 3. The solution was found to be $x = 0.739$.

4.10.2 Example Problem 2: Numerical Integration

Problem Statement: Compute the integral of $f(x) = x^2 \sin(x)$ from 0 to π using the Simpson's rule.

Solution: NumLab was used to perform the integration using Simpson's rule. The result obtained was accurate to within 10^{-6} of the analytical solution.

4.10.3 Analysis of Results

The results from these case studies demonstrate the accuracy and reliability of NumLab in solving complex numerical problems. The solutions obtained were consistent with theoretical expectations and matched well with results from other established tools.

4.10.4 Discussion on the Application's Utility

These examples highlight the practical utility of NumLab for students, educators, and professionals in the field of numerical analysis. By providing accurate solutions and visualizations, NumLab proves to be an invaluable tool for learning and applying numerical methods.

Conclusion and Future Work

5.1 Summary of Achievements

NumLab successfully implements a range of numerical methods, providing users with a powerful tool for solving complex problems in numerical analysis. The application is user-friendly, accurate, and versatile, making it suitable for a wide audience.

5.2 Evaluation of Results

The results obtained through NumLab are validated against benchmark solutions, confirming the accuracy and reliability of the methods implemented. The application's performance is also competitive, handling complex calculations efficiently.

5.3 Contributions to the Field

NumLab contributes to the field by offering an accessible platform for learning and applying numerical methods. It bridges the gap between theoretical concepts and practical applications, making numerical analysis more approachable.

5.4 Limitations and Challenges

Despite its strengths, NumLab has some limitations, including the handling of highly complex problems that may require more advanced techniques or specialized software. Future work could address these limitations by incorporating more sophisticated algorithms and expanding the range of problems that NumLab can solve.

5.5 Future Work and Potential Enhancements

Future enhancements for NumLab could include the addition of more numerical methods, improved visualization tools, and the integration of machine learning techniques to further enhance its capabilities. Additionally, expanding the user base and collecting more feedback will help guide the development of new features and improvements.

5.6 Conclusion

The development of NumLab has demonstrated the feasibility of creating a web-based platform for solving numerical analysis problems. The project not only provides a valuable tool for learning and applying numerical methods but also serves as a foundation for further development and research in the field. The positive feedback from users suggests that NumLab has the potential to become a widely used educational resource, and future enhancements could expand its capabilities even further.

Bibliography

Flask documentation. <https://flask.palletsprojects.com/en/2.0.x/>. Accessed: 2024-08-25.

Mathjax documentation. <https://docs.mathjax.org/en/latest/>. Accessed: 2024-08-25.

atkinson. *sec₃ – 3.dvi*.

Kendall E. Atkinson. *An Introduction to Numerical Analysis*. Wiley, 1989.

Richard L. Burden and J. Douglas Faires. *Numerical Analysis*. Cengage Learning, Boston, MA, 9th edition, 2010. ISBN 978-0-538-73351-9.

James F. Epperson. *An Introduction to Numerical Analysis*. John Wiley & Sons, 2007.

Walter Gautschi. *Numerical Analysis*. Birkhäuser, Basel, 2nd edition, 2001.

Amos Gilat and Vish Subramaniam. *Numerical Methods for Engineers and Scientists*. John Wiley & Sons, 2008.

David Kincaid and Ward Cheney. *Numerical Analysis: Mathematics of Scientific Computing*. Brooks/Cole, Cengage Learning, 3rd edition, 2009.

Sir Onassis. Solving algebraic and transcendental equations - part i.xopp. a.

Sir Onassis. Solving algebraic and transcendental equations - part ii.xopp. b.
fla, mat, atkinson, Atkinson [1989], Burden and Faires [2010], Epperson [2007], Gautschi [2001], Gilat and Subramaniam [2008], Kincaid and Cheney [2009], Onassis [a,b]